# 转发：JBHI-06328-2025

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Reject/Resubmit (major revision and new external review required)

Dear Prof. Xinhua ZENG,

This letter is to inform you that the peer review process has concluded for manuscript, "PathVLM-R1: A Reinforcement Learning-Driven Reasoning Model for Pathology Visual-Language Tasks," which you had submitted for possible publication in the IEEE Journal of Biomedical and Health Informatics (J-BHI).

The Associate Editor responsible for your manuscript review has received feedback from independent reviewers and compiled their evaluations. It is the recommendation of the Associate Editor and the Editor-in-Chief that your manuscript requires a MAJOR REVISION before it can be accepted for publication in J-BHI. Please note the J-BHI Editorial Policy that only one major revision is allowed for any submitted manuscript. This means that if the review recommendation for your revised paper triggers another major revision, the paper will be rejected automatically.

Enclosed, please find the comments by the Associate Editor and all the reviewers. I hope that the feedback is helpful for further improving the quality of your manuscript. If you decide to resubmit a revised manuscript, this must be done within 10 weeks (not extendable) from the date of this message. Please quote the above manuscript reference number for all future correspondence.

***IMPORTANT: PLEASE RETURN THE IEEE JBHI AUTHOR PORTAL TO SUBMIT ALL FILES: https://ieee.submission.researchexchange.com/journal/jbhi-embs**

Also note that it is mandatory to enter your replies to the reviewers' questions and indicate how you have dealt with their comments in the revised manuscript. Please include your replies in the authors' response section or a separate file with a point-by-point explanation of all the changes made. Please do NOT include it in the cover letter since this is not accessible by the reviewers. For submitting your revised manuscript, please ensure all changes are highlighted in the manuscript to facilitate the review process.

Citing/including papers suggested by the reviewers or the Associate Editor is up to the authors and only if they add value to your work. Please also report to ontolimatics@gmail.com any suspicious suggestion by the reviewers.

The authors are responsible to follow the J-BHI publication rules for maximum number of pages, quality of figures, etc. as they are mentioned in:
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On the opposite your article might need to be returned to you and the review process will be delayed and in case of acceptance no publication is possible.

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Thank you very much for considering JBHI to publish your research work.

Sincerely,

Prof. Peter Elkin
Editor-in-Chief

Cc: file

Associate Editor's comments to the authors:
Associate Editor
Comments to the Author:
The Reviewers agree that this paper provides some interesting ideas. Howevr, there are several open questions and weaknesses that should be carefully addressed before a new round of reviews.

For submitting your revised manuscript, please ensure all changes are clearly highlighted in the manuscript and explained in detail in the rebuttal to facilitate the review process.

Reviewers' comments to the authors:
Reviewer: 1

Comments to the Corresponding Author
This paper proposes PathVLM-R1, a visual-language model specifically designed for pathological image analysis. The authors' motivation to address the "black-box" nature of medical visual-language models aligns well with the scope of JBHI. By adapting concepts from reasoning models like DeepSeek-R1 and DeepSeek-Math to pathology image analysis, the authors have achieved good results, demonstrating value in applied innovation.

However, there are several methodological and presentation issues that need to be addressed.

Concerns:
1. A foundational premise of this study relies on using GPT-4o as a surrogate medical expert to establish the reward mechanism. Although Figure 4 demonstrates the consistency of GPT-4o's evaluations across multiple trials, this consistency does not inherently prove clinical accuracy. While the authors acknowledge this limitation, the paper would be significantly strengthened if a small-scale validation study were conducted comparing GPT-4o's reward scores with those of human experts, thereby justifying the validity of this approach.

2. It is highly important to include an analysis of typical failure cases of the model (e.g., when diagnosing rare diseases). This will help to clearly articulate the model's current limitations and outline concrete directions for future improvement.

3.The authors repeatedly emphasize the model's "remarkable cross-modal transferability." However, while its performance on the ISIC 2020 dataset is impressive (80.00%), its accuracy on Chest CT and Retinal OCT-C8 is actually lower than that of the baseline model, Qwen2.5-VL-7B (as shown in Table IV). The authors must provide a thorough explanation for this performance degradation in certain modalities.

4. In the methodology section describing the reward mechanism, the authors state that 0.4 points are deducted for each missing reasoning step. What is the empirical basis or rationale for choosing this specific penalty value? Please clarify.

5. Regarding the evaluation criteria, it is noted that GPT-4o is used both to construct the reward mechanism during the reinforcement learning phase and later as the evaluator for the final model performance. This dual role creates a circular evaluation loop, rendering the current evaluation system insufficiently objective and rigorous.

6. Figure 4 is too small, making the text and details difficult to read. Please improve its resolution and enlarge it for better legibility.

7. In Table II, please explicitly specify the experimental configurations or prompting strategies used for the "Large-parameter models" and "Comparable-parameter models" to ensure a fair and transparent comparison.

8. The attention heatmap in Figure 7 lacks a color scale bar, making it impossible for readers to quantify the intensity of the attention weights accurately.

9.There is a noticeable mix of terminology throughout the manuscript. For example, "PathVLM-α" and "PathVLM-Alpha", as well as "procedural loss" and "process reward", are used interchangeably. Please thoroughly proofread the manuscript and standardize these terms globally.

Reviewer: 2

Comments to the Corresponding Author
This manuscript proposes PathVLM-R1, a pathology visual-language model built on Qwen2.5-VL-7B-Instruct and optimized through staged supervised fine-tuning and GRPO-based reinforcement learning. The main technical idea is a dual-reward mechanism combining answer/format rewards with a GPT-4o-based cross-modal process reward for evaluating the generated reasoning chain. The topic is timely, and the staged comparison between SFT, outcome-only RL, and process-aware RL is a useful direction.

However, I have several major concerns.

First, the experimental setting is too limited for the strength of the claims. The study mainly uses a small subset of PathMMU, with 3,000 samples for SFT, 1,000 samples for RL, and 1,385 samples for testing/validation. This is a small data regime for adapting a 7B VLM, especially with many optimization steps. The paper should avoid claims of a broadly generalizable or foundational pathology model unless substantially stronger evidence is provided.

Second, the train/test split is not sufficiently specified. Since PathMMU may contain multiple Q&A pairs per image, it is important to clarify whether the split was performed at the Q&A-entry level, image level, or case/patient level. If the same image appears in both training/RL and testing with different questions, the reported performance may overestimate generalization. Please provide image-level and case-level de-duplication details across SFT, RL, validation, and test sets.

Third, the baseline comparison is incomplete. The manuscript compares with several general or medical VLMs, but does not include enough pathology-specific VLM/LMM baselines, including strong non-RL pathology models. Since the paper claims superior performance on pathology visual-language tasks, comparison with representative pathology-specialized models is necessary. Otherwise, it is difficult to determine whether the gain comes from the proposed dual-reward RL strategy or from comparing against less domain-adapted baselines.

Fourth, the GPT-4o reward mechanism needs more detail and validation. Please clarify whether GPT-4o rewards are computed online during RL or precomputed/cached offline, and report the exact prompts, model version, decoding parameters, parsing/fallback rules, number of calls, and cost. Since GPT-4o is given the standard answer, the process reward may partly evaluate answer-consistent rationalization rather than faithful visual reasoning. Human pathologist validation of the reward scores would strengthen the paper.

Finally, the evaluation scope remains narrow. The current evaluation is mainly multiple-choice pathology VQA on a subset of PathMMU. Additional evaluations on official PathMMU splits and independent pathology benchmarks would make the empirical conclusions more convincing. The paper should also report results on the full held-out set rather than only a 500-sample comparison, and provide confidence intervals or repeated-run statistics.

Overall, the dual-reward idea is potentially interesting, but the current manuscript requires substantial additional experiments, clearer data-splitting details, stronger pathology-specific comparisons, and more cautious claims.

Reviewer: 3

Comments to the Corresponding Author
This manuscript presents PathVLM-R1, a pathology visual-language model built on Qwen2.5-VL-7B-Instruct. The method uses a staged post-training pipeline: supervised fine-tuning for pathology knowledge injection, followed by reinforcement learning with GRPO using outcome rewards and a GPT-4o-based cross-modal process reward. The authors report improved in-domain performance on PathMMU pathology VQA tasks and argue that the model provides more interpretable reasoning and cross-modal generalization to other medical imaging modalities.

While the paper addressing an important problem that attempted to improve reasoning transparency and diagnostic accuracy in pathology-oriented LLMs, several concerns have been raised from my side, they are:

1. The claim of cross-modal transferability is not supported by the reported results. The manuscript claims strong cross-modal transferability, but Table IV does not support this conclusion. Compared with Qwen2.5-VL-7B, PathVLM-R1 improves only on ISIC 2020, from 56.59% to 80.00%. However, it performs worse on the other three out-of-domain datasets: Chest CT decreases from 48.60% to 44.00%, Retinal OCT-C8 decreases from 57.19% to 49.20%, and Diabetic Retinopathy decreases from 70.80% to 59.80%. Therefore, the evidence does not justify a general claim of cross-modal transferability. The authors should either substantially tone down this claim or provide stronger out-of-domain experiments, modality-wise analysis, and statistical validation.

2. The manuscript uses a single attention heatmap in Fig. 7 to support visual grounding and states that it provides clear evidence of interpretability and enhances trustworthiness. However, this is only a qualitative showcase. A single example cannot substantiate broad claims about interpretability or trustworthiness. Moreover, attention maps do not necessarily provide faithful explanations of model decisions. The authors should add quantitative grounding or faithfulness evaluations, such as expert-rated localization, region-overlap metrics when annotations are available, counterfactual image perturbation, deletion/insertion tests, or evaluation across multiple representative cases.

3. The overall pipeline: supervised fine-tuning followed by GRPO-based reinforcement learning with format, accuracy, and GPT-based process rewards, is a relatively standard recent reasoning-model training strategy. The paper should more clearly explain what is genuinely novel beyond applying this existing pipeline to pathology VQA. If the key contribution is the GPT-4o-based cross-modal process reward, the authors need stronger ablations showing that this component provides a clear and statistically meaningful improvement over SFT-only, outcome-reward-only, and simpler rule-based or model-based reward designs.

4. The manuscript should provide a more detailed discussion of prominent pathology foundation-model and pathology VLM work, especially the line of work represented by UNI, CONCH, and PathChat from Mahmood Lab. Even if direct performance comparison is not feasible, the paper should clearly position PathVLM-R1 relative to these models in terms of task scope, training data, supervision strategy, model capability, clinical use case, and limitation.

5. The paper emphasizes that PathVLM-R1 can achieve strong performance with only 3,000 SFT samples and 1,000 RL samples. However, there is no systematic data scaling experiment. The authors should report how performance changes with different SFT and RL data sizes, for example 500, 1,000, 2,000, and 3,000 SFT samples, as well as different RL sample budgets. This would help determine whether the reported gains are robust or dependent on a particular data size and split.

6. The results are mainly presented as point estimates, without confidence intervals, significance tests, or repeated runs with different random seeds. Given the relatively limited test size and the sensitivity of VLMs to prompting and parsing rules, the authors should report variability across runs and provide statistical evidence that the improvements are reliable.

7. GPT-4o is central to the proposed reward mechanism, and GPT-based scores are also used to evaluate reasoning quality. This may introduce circularity, because the model is partly optimized toward the same kind of judgment used for evaluation. The authors should validate the GPT-based reward against human pathologist assessments or use independent evaluator models and report agreement with expert judgments.

8. There are typographical and formatting errors. For example, “Doubao-1.5-vison” should likely be “Doubao-1.5-vision.” The manuscript would benefit from careful proofreading for grammar, formatting, and consistency.